

3 (a) Any three of

[3]

- Intermolecular forces are negligible except during a collision.
- All collisions are perfectly elastic.
- The duration of a collision is negligible compared with time interval between collisions.
- The gas molecules are in rapid, random motion and obey Newton's laws of motion.
- Gas consists of large number of molecules.
- The volume of the all the gas molecules themselves is negligible compared with the volume occupied by the gas. (In other words, the gas consists of mostly empty space)

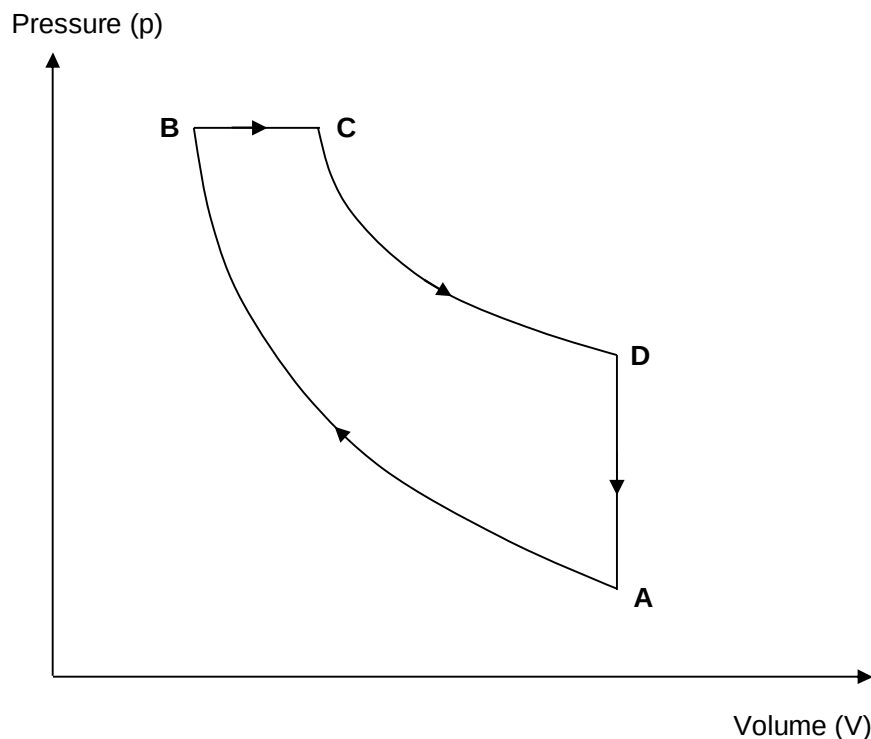
(b)
$$n = \frac{PV}{RT} = \frac{(1.0 \times 10^5)(6.0 \times 10^{-4})}{(8.31)(300)} \quad [1]$$

$$= 0.024 \quad [1]$$

$$N = nN_A = (0.024)(6.02 \times 10^{23}) \quad [1]$$

$$= 1.45 \times 10^{22}$$

(c)



[First 3 marks given for purely 3 correct lines (no need arrow / and steepness of AB and CD)]

[Last mark for correct line + arrow + correct steepness of curve]